

Interactive 3D Experience Diploma in Immersive Media & Game Design Year 2 (2025/26) Semester 2.1	Version 1
	Date Apr 2026
Assignment Project 1 (30%) Creation of Basic 3D Virtual Player Experience	Week 3 - 8

OVERVIEW

REQUIREMENTS

- Creation of a basic 3D virtual player experience.
- This is an **individual** project

INTENDED LEARNING OUTCOMES

1. Utilize C# programming language and Unity engine to create a basic 3D virtual world
2. Usage of C# to create logical workflows and gameplay logic
3. Usage of Unity Physics, Animation, and Audio to create gameplay.
4. Updating and displaying progress through Unity UI.
5. Usage of Git to track progress and maintain project.

1. INTRODUCTION

The Unity Engine allows developers and designers to create tangible, meaningful interactive experiences from their imaginations. It offers a wide array of tools that enable its users to create both 2D and 3D applications.

For this assignment, you will assume the role of 3D Unity Developer and use your skills to create a basic player experience. You are tasked to create a 3D virtual environment that contains collectibles that users can collect, and hazards and obstacles that damage and obstruct the user. The user must control an avatar through a first-person camera view.

The target platform for this project is Windows desktop. You are required to test all assets, scenes, and scripts are working to your needs.

2. SUBMISSION CRITERIA

All files are to be submitted to BrightSpace.

Submit all deliverables individually and NOT collated as a zip file.

Note: Always backup your files using other mediums (eg. Google Drive, Dropbox, External HDD, ictspace server) throughout the assignment.

No.	Deliverable	Naming Convention	File Format
1.	Git Repository to Unity Project	-	-
2.	Read Me	I3E_Asg1_<Name>_ReadMe.pdf	.pdf
3.	Gameplay & Code Presentation	I3E_Asg1_<Name>_Gameplay.mp4	.mp4

3. DELIVERABLE DETAILS

1. Git Repository to Unity Project

- Link to be submitted via BrightSpace as a **comment during submission**.
- Repository should be set to **Public**.
- Include all project files (e.g scenes, scripts, materials, textures, packages, models, etc)
- All files should be organized in a proper file structure format

*Note: Files should be properly named and structured. Failure to do so will result **in heavy penalization** of marks. Marks are given for proper file organization.*

2. ReadMe

- Readme should include detailed instructions (key controls, game hack, answer key, etc) on how to use and run your application
- Indicate the **platforms/hardware/desired settings** required to run your application
- Indicate the limitations or bugs in the application
- Indicate **references and credits to the models/textures/materials** used
- Include the answers on how to solve your game puzzles if any

3. Gameplay Video

- Video of gameplay from start to end of the experience.
- Elaborate on the development process after the gameplay showcase.
 - Elaborate on:
 - Code and logic
 - Unity systems used (e.g. Physics, Animation, etc.)
 - Set up of scene and objects
- Video length between 5 to 10 minutes long.

4. IMPORTANT TASKS & DELIVERABLES

- Read the requirements for the application carefully. Some requirements are **necessary for a Pass grade**.
- For submission, you should
 - Upload the files to the assignment submission link in **BrightSpace**
 - Cross check all your files are working and can be opened. It is not uncommon that students submit shortcuts instead of the working files.
- All documents submitted must be in **PDF format**

DUE DATE

Assignment is due on **Week 8**.

TECHNICAL REQUIREMENTS

Your development project is to **include the following**:

- Usage of GitHub Desktop to manage the project.
- Interactions with collectibles must be done with Unity Physics.
- Utilization of XML comments (“///”) for ALL variables and functions.
- Utilization of file headers for ALL script files.

File header template:

```
/*  
* Author:  
* Date:  
* Description:  
*/
```

5. ASSIGNMENT WEIGHTAGE

ITEMS	WEIGHTAGE
Scripting & Project Management <i>Are best practices being adhered? Have XML comments and file headers been used? Is the project properly organized? Are prefabs being used when appropriate? Is code structured and logical?</i>	40%
Interactivity <i>How much interactivity is there between player and the environment? Are there interactions beyond colliding with objects? Are there more than one type of collectible? How many hazards are there?</i>	30%
UI/UX <i>Does the on-screen UI update when the player collects items? Is there a message that congratulates the player on collecting all items?</i>	15%
Game Juice <i>Are proper sound effects and BGM being used? Are there different sound effects for different events? (e.g. collecting sfx, damage sfx)</i>	15%
GRAND TOTAL	100%

MARKING SCHEME

To obtain this grade...	Minimally, you need to implement...
Pass	• Proper maintenance of project on GitHub • Collisions via Unity Physics • Player and environment models • 2 areas separated by doors • Doors open when the player is near them. • At least one type of hazard that kills the player • At least 5 collectibles ○ Can be duplicated objects ○ Collectibles are collected when the player touches it • UI displaying the current player score • UI congratulating the player on collecting all 5 collectibles • UI displaying a game over screen when the player is killed. • Script files are properly commented with file headers, and line and XML comments • BGM is implemented
Merit Grade	You must fulfil the criteria of obtaining a passing grade, plus: • Collectibles are collected when the player is near the collectible and presses an “Interact” button. • Score of each collectible can be adjusted in the Unity Editor • UI displays how many collectibles are left • Doors are locked until the player fulfils some criteria. (e.g. collecting a special collectible, passing through an area, etc.) • More than one type of hazard. • Hazards deal damage over time to the player instead of kill the player on one hit. • Interactions have SFX implemented.
Excellent Grade	You must fulfil the criteria of obtaining a Merit grade, plus: • Multiple types of unique collectibles. • All interactions are done using Raycasting • Doors close when the player moves away from them. • Doors are opened when the player presses an “Interact” button. • Interactions are done using Raycasting and the “Interact” button • Different types of hazards deal different amounts of damage to the player. • Multiple access points for the 2 areas

**Work from previous classes/other modules CAN be used for this assignment. But it is up to you to make sure it fits into the requirements.*

**3D assets built using ProBuilder will also count as custom assets.*

6. SUBMISSION DEADLINE

Deadline: 2359, Sunday of Week 8 (14/06/2026)

All deliverables to be properly named and submitted on **BrightSpace**.

CODE REVIEW

As part of your assessment, you may be required to give a brief explanation of selected pieces of code written by you in the assignment.

PLAGIARISM AND COPYRIGHT

Plagiarism means, “*copying any part of a source, and then submitting it, claiming that it is your own work.*”

Please ensure that all the works submitted by you are not copied from other sources. Any attempt to plagiarize will be dealt with severely, and it may result in your failing the module.

- This is an **individual** assignment. Do not copy code from your classmates or online sites. All copied code must be attributed and credited
- Requirements implemented with the help of third party code will not be counted as implemented by you, and may result in loss of marks.
- If you are found committing plagiarism, you will score 0 for your assignment

LATE SUBMISSION

Late submission will be **penalised** (10% of the marks for each day late after 12 noon). Submission will **not be accepted after 5 days** (including weekends and public holidays) from the date of submission.

DUE DATE

Assignment is due on **Week 8, Sunday @1159pm.**

- End of Project Brief (Total: 10 pages) -